OSINT Shield — technical report

Classifying defence & security news with a fine-tuned multilingual encoder, and running it live. November 2026 · code: this repository · decisions log: DECISIONS.md (D1–D25) · day-by-day log: PROGRESS.md

Summary

- **Task.** Label defence and security news articles with a **narrative** (Security, Political, Civilian, Investigation, Other), a **severity** (High / Low) and **propaganda** (yes / no), from 1,064 articles labelled by two LLM annotators (Claude, Llama).
- **Headline result.** A fine-tuned `mmBERT-small` (42 M parameters in the encoder body), scored once on 94 held-out articles, reaches narrative macro-F1 **0.415 ± 0.014**, 3-class **0.602 ± 0.039**, severity F1(High) **0.664 ± 0.045** — **statistically indistinguishable from TF-IDF + logistic regression** (every paired 95% bootstrap interval includes zero). Its cross-validation lead (+0.057) did not survive contact with unseen data.
- **Why.** Label quality is the ceiling, not the model: the two annotators agreed on 74.3% of narratives, and the model is **85.3%** accurate where they agreed versus **50.0%** where they did not. 56% of its errors fall on the 28% of articles the annotators disagreed on.
- **What else was measured, not assumed.** Truncating at 512 tokens costs nothing measurable; keyword fusion helps a little, mainly on the rare Investigation class; headline-only articles are barely classifiable and the model is *over*-confident on them; propaganda (7 labelled positives) cannot be learned and is reported as *not reliably assessed*.
- **Live system.** RSS + Google News collection every 15 minutes, a relevance filter, article-text extraction, the classifier, and Llama 3.2 3B writing summaries whose quotes are **checked word for word** (37 invented quotes caught in the first hour), behind a FastAPI service and a React dashboard — all on one 6 GB laptop GPU.
- **Biggest open problems.** Label adjudication; what "High severity" means (in this corpus it often means *militarily significant topic*, not *incident now*); and article text for Google News items, which an undocumented Google endpoint provides only until it rate-limits.

1 · Data

fact	value	consequence
articles	1,064	small: every estimate needs its uncertainty
narrative classes	Security 824 (77.4%) · Political 155 · Civilian 41 · Other 26 · Investigation 18 · Radicalization 0	macro-F1, not accuracy; Radicalization dropped (D6)
severity	High 241 (22.7%) · Low 823	F1 on the High class
propaganda	7 positives	diagnostic only (D3)
no body text	490 (46%), all Google News	the main data-quality issue
body ⇄ severity confound	High: 34.0% of bodied vs 9.5% of headline-only	a classifier fed *only* "has a body?" scores F1(High) 0.467
annotator agreement	narrative 74.3% · severity 79.9%	a ceiling on any honest score
duplicates	11 exact; 22 near-duplicate clusters, 12 spanning the original splits	group-aware folds (D5)

fact	value	consequence
length	bodied median 3,618 chars; 74.8% of inputs truncated at 512 tokens	tested in D14
language	English 1,032 · Hindi 29 · Kannada 3	multilingual claims are untestable

Evaluation protocol (D1, D2, D5). The original `test.csv` (159 articles) was locked away on day one and scored **once**, at the end. Development used 5-fold cross-validation over the remaining articles that have a body (476), with near-duplicates grouped so no story sits in both training and test, propaganda positives stratified, and 3 seeds per fold (15 runs per configuration). Training uses bodied articles only (`body_only`): including headline-only rows would let the model learn the body ⇄ severity shortcut. Details: DATA_REPORT.md.

2 · Model

```
headline <eos> body ──► mmBERT-small encoder (frozen word embeddings, SDPA attention)
   (512 tokens)                │ mean pooling
                               ▼
     11 rubric keyword-group counts (whole article, log1p, standardised) ──► concat
                               ├──► narrative head  (5 classes)
                               ├──► severity head   (2 classes)
                               └──► propaganda head (2 classes, loss weight 0.3, pos. weight ≤ 10×)
```

setting	value	why
encoder	`jhu-clsp/mmBERT-small`, **mean pooling**	its own config's pooling; `[CLS]` pooling cost 0.14 macro-F1 (D11)
memory	frozen word embeddings + SDPA attention	4.73 → 2.63 GB, fits a display-sharing 6 GB GPU (D9)
keyword fusion	rubric group counts over the full text	user directive (D4); small measured gain (D12)
schedule	fixed 6 epochs, AdamW 2e-5, warmup 10%, linear decay	early stopping on ~47 rows was noise (D11, D13)
batch	8 × accumulation 2, bf16 autocast	
chosen by	a rule fixed **before** the deciding run (D15)	

Code: `models/`, `training/`, configs `base.yaml` → `mmbert_small.yaml`.

3 · Results

3.1 Baselines — the bar to clear

Cross-validation, bodied articles, mean ± std over 5 folds.

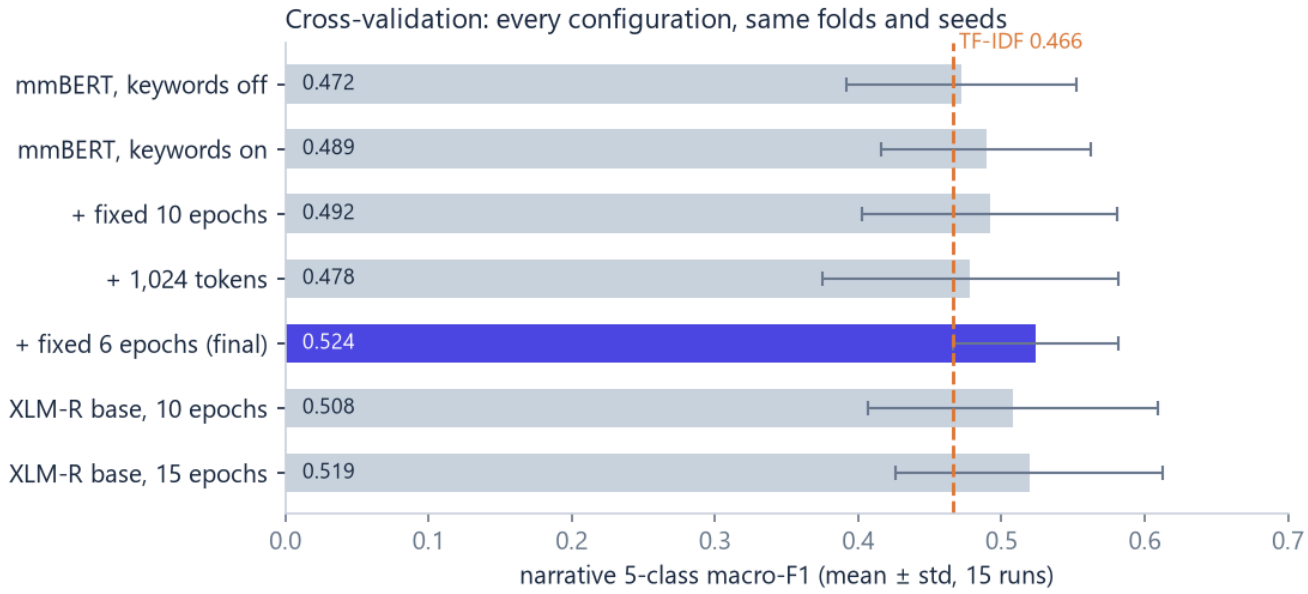
method	narrative 5-class	narrative 3-class	severity F1(High)	propaganda F1
majority class	0.172	0.287	0.000	0.000
keyword rubric (rules)	0.443 ± 0.058	0.517 ± 0.060	0.667 ± 0.038	**0.118**
keyword counts + logistic regression	0.383 ± 0.033	0.474 ± 0.032	0.675 ± 0.039	0.063
TF-IDF + logistic regression	**0.466 ± 0.015**	**0.604 ± 0.038**	**0.687 ± 0.072**	0.000

Without duplicate grouping, TF-IDF's 3-class score reads 0.627 instead of 0.604 — leakage that grouping removes. The hand-written rubric is a strong severity baseline and the only method with any propaganda signal, because its propaganda terms were copied from the positives (D18, D20).

3.2 Model development — what each experiment decided

Same 5 folds × 3 seeds for every row; mean ± std over the 15 runs.

configuration	narrative 5-class	narrative 3-class	severity F1(High)	train time
mmBERT, `[CLS]` pooling (M5 v1)	0.330	—	—	—
mmBERT, mean pooling, keywords off	0.472 ± 0.080	0.616 ± 0.061	0.656 ± 0.047	28 min
mmBERT, mean pooling, keywords on	0.489 ± 0.073	0.625 ± 0.049	0.666 ± 0.042	29 min
+ fixed 10 epochs (no early stopping)	0.492 ± 0.089	0.629 ± 0.053	0.677 ± 0.069	35 min
+ 1,024 tokens	0.478 ± 0.103	0.612 ± 0.064	0.688 ± 0.059	70 min
+ fixed 6 epochs — final	**0.524 ± 0.058**	**0.646 ± 0.038**	**0.692 ± 0.060**	20 min
XLM-R base, 10 epochs	0.508 ± 0.101	0.642 ± 0.055	0.678 ± 0.082	44 min
XLM-R base, 15 epochs	0.519 ± 0.093	0.648 ± 0.055	0.656 ± 0.059	61 min



What the paired comparisons (same fold, same seed) decided:

question	answer	decision
Why was the first run *below* TF-IDF?	three defects: `[CLS]` pooling on a model built for mean pooling (2 of 3 layers see only a 128-token window), an early-stopping signal an all-"High" model could game, and selection on 47 rows	fixed; +0.14 from pooling alone (D11)
Do keywords help?	+0.017 ± 0.072, 10/15 runs; Investigation F1 0.329 vs 0.254	keep, as "directionally positive, not established" (D12)
Early stopping or a fixed schedule?	a tie on score; the final all-data model has nothing to early-stop on	fixed (D13)
Is truncation at 512 tokens hurting?	1,024 tokens: −0.014 ± 0.101, 6/15; twice the time	keep 512 — the tail carries no measurable signal (D14)
mmBERT-small or XLM-R base?	narrative tie (−0.004, 6/15); mmBERT wins severity in 12/15 runs, steadier, half the size	mmBERT-small (D15)

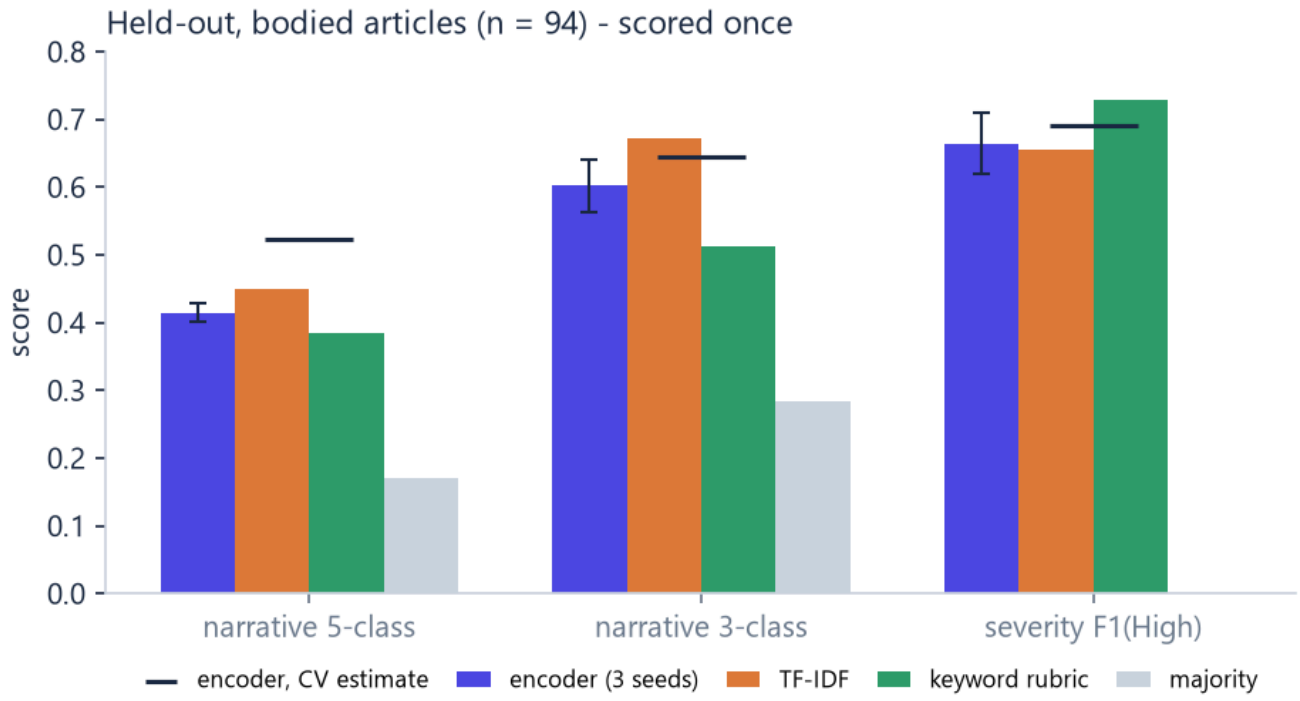
The final recipe beat TF-IDF on narrative in 4 of 5 folds (+0.057). It was, however, the best of ~8 configurations scored on the same folds, so that lead was expected to shrink.

3.3 Held-out — the headline number

Scored once, 2 October 2026, on the 94 bodied articles of `test.csv` untouched since day one. Encoder = mean ± std over 3 seeds; the deployed model is seed 42, fixed in advance.

task	encoder	TF-IDF	keyword rubric	majority	CV estimate
narrative 5-class	**0.415 ± 0.014**	0.449	0.385	0.171	0.524
narrative 3-class	**0.602 ± 0.039**	0.673	0.513	0.285	0.646
severity F1(High)	**0.664 ± 0.045**	0.656	0.730	0.000	0.692

Paired bootstrap (4,000 resamples), encoder − TF-IDF: narrative **−0.023 [−0.181, +0.053]**, 3-class **−0.040 [−0.133, +0.064]**, severity **+0.039 [−0.087, +0.171]**. All include zero.



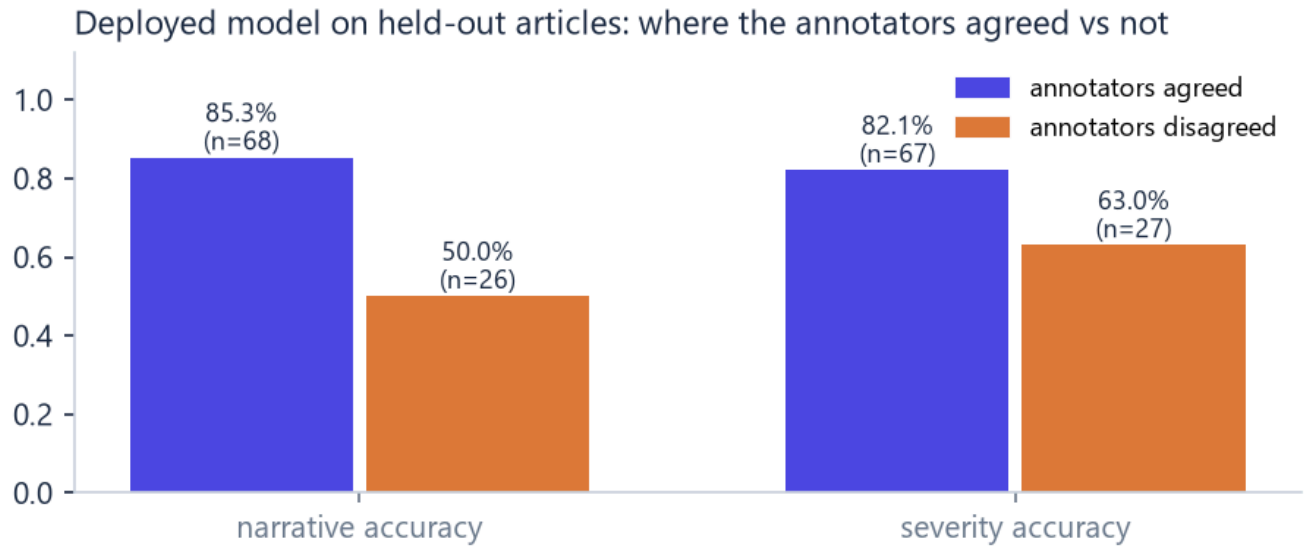
Reading it.

1. **Encoder and TF-IDF are statistically indistinguishable.** The CV advantage did not replicate.
2. **The 0.109 drop from the CV estimate is the expected winner's curse** (best of ~8 on the same folds) plus a tiny rare-class sample: the held-out set has **2 Investigation and 2 Other** articles, each worth ~0.1 of 5-class macro-F1. All three seeds dropped alike (0.426 / 0.423 / 0.395).
3. **The number is frozen.** Changing anything in response to it would be tuning on the test set.

Per-class F1 (deployed model): Security 0.853, Civilian 0.667, Political 0.611, Investigation 0.000, Other 0.000. Full tables: FINAL_RESULTS.md.

3.4 Error analysis — the ceiling is the labels

held-out bodied articles	n	narrative accuracy	severity accuracy
annotators **agreed**	68 / 67	**85.3%**	82.1%
annotators **disagreed**	26 / 27	**50.0%**	63.0%



56% of narrative errors fall on articles the annotators disagreed on — twice their 28% share. Where the "truth" is itself a coin-flip between two annotators, no model can score well, and a higher-capacity model has nothing more to learn. **More or cleaner labels would move the numbers; a bigger model would not** (D16).

3.5 Headline-only articles

Evaluated on the 415 development headline-only articles the bodied model never trained on — not on `test.csv` (D17).

narrative	5-class	3-class
encoder	0.209	0.348
keyword rubric	0.208	0.370
hybrid — rubric when a narrative term fires, else encoder	**0.235**	**0.413**

The hybrid was adopted under a rule fixed in advance (switch only at paired-bootstrap P ≥ 0.90; it reached 0.98). Three findings matter more than the routing choice:

- **A headline is barely enough.** A majority-class predictor scores ~0.18; the best method 0.235. The fix is engineering — fetch the article — not modelling.
- **The model is over-confident where it is weakest:** mean narrative confidence 0.872 on headlines versus 0.823 with a body. Confidence-based escalation would wave headline errors through, so every headline-only article is routed to review **by type**, regardless of confidence.
- The 65 held-out headline-only articles had suggested the rubric was far better (0.528 vs 0.276). On 415 it is not. The pre-registered rule kept a small-sample artefact out of production.

3.6 Propaganda — measured, then set aside

With 7 positives, every learned method scores F1 = 0. A re-annotation was run (D18–D20):

- **Local LLMs cannot pre-screen it.** Llama 3.2 3B and Qwen 2.5 7B, across three prompt designs, could not tell an outlet's own voice from an attributed quote — the core of the annotation rule. They filled the output schema without performing the judgement it encodes.
- **100 articles reviewed blind** (7 originals, 53 rubric hits, 40 random): 9 positives, 5 of them sure — all 5 originals. Rubric precision **6.9%**; unflagged articles carried propaganda at 7.5%, rubric-flagged at 1.9% — the rubric finds the articles it was copied from, nothing more.
- A trainable set (~50 positives) would need ~600 more random articles reviewed. The live system therefore reports propaganda as **"not reliably assessed"** rather than a guess.

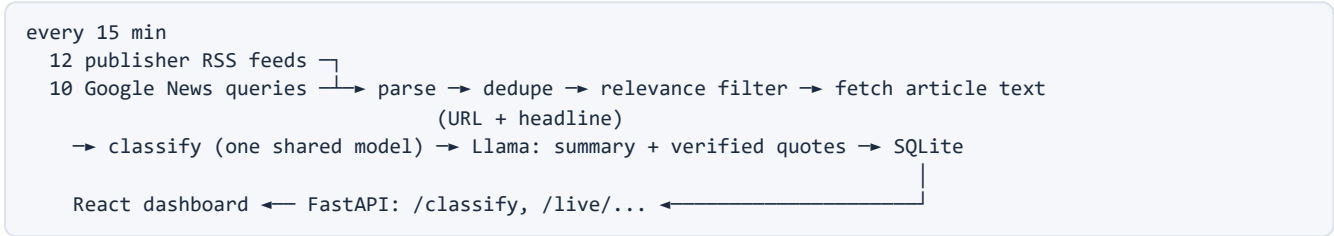
3.7 Serving

Measured on 1,428 out-of-fold development predictions and the deployed model (D21).

calibration (ECE)	narrative 0.036 · severity 0.040 — confidence is trustworthy on bodied articles
escalation at confidence < 0.70 or High	59% of articles escalated, catching 58% of narrative errors; accuracy 87.3% above the threshold vs 49.5% below
latency (fp32)	**26.8 ms/article batched**; 48.0 / 72.3 ms p50 / p95 for single calls
consistency	0 / 160 label differences between an article scored alone and in a batch — bf16 had 1 / 160, so inference runs in fp32
version stamp	`mmbert-small-s42-b675e23f` on every record

The plan expected ~25% escalation; it assumed the corpus-wide High rate (22.7%), while bodied articles are 34% High. At ~4 s per Llama call and a few hundred articles a day, 59% costs about 12 GPU-minutes a day.

4 · The live system



stage	rule	evidence
sources	the outlets the training data came from, publisher feeds first	489 of 1,064 gold articles came through Google News (D22)
dedupe	canonical URL hash + normalised headline; a later copy with text upgrades a headline-only one	a second pass stores nothing new
relevance filter	a defence term in the headline, or ≥ 2 distinct terms in the text	99.2% recall on the gold corpus (rubric alone: 71%); admits ~39% of general-news items (D22)
article text	trafilatura on the publisher page, ≥ 400 chars, headline must match the page	defence.in, Indian Express, The Hindu, HT, Livemint, BusinessLine, TOI extract; The Diplomat and SCMP refuse automated clients (D23)
Google News links	2-step decode through an undocumented Google endpoint	12/12 correct, then HTTP 429 after ~40 requests → ≤ 10 per cycle, 6 s apart, 1-hour pause on 429
classify	M9 routing; waits for text, at most 60 min	
explain	escalated articles **with text**, High first; Llama gets no labels, returns quotes + summary	quotes not found in the article are dropped (D23)
alerts	High **and** has text **and** not an analysis outlet	see below (D23)
concurrency	one cycle at a time across processes (database lease); one shared model behind a lock	D24

First hour of operation (1 November 2026, 5 cycles): 1,050 items stored, 473 relevant, 423 classified; 89 Llama explanations at 3.7–4.3 s each, 87 with verified quotes, **37 invented quotes caught and dropped**; steady-state cycles 39–62 s; GPU 4.3 of 6 GB with the classifier, Llama and the display.

What "High" means — a finding the live data forced. Of the first 29 bodied High labels, 22 came from 19FortyFive feature pieces ("Patton's best Army line", "the F-16N summed up in one word"), while a gunfight that killed three people in Manipur sat at P(High) 0.50. The model is faithful to its labels: in the training data **42% of**

19FortyFive and 44% of Naval News articles are High. The annotators' "High" often means *militarily significant topic*. This cannot be fixed without relabelling, so the dashboard separates analysis outlets and only raises **alerts** for High articles that have text and come from news outlets: the first alert list became a Patriot-missile deal, 22 Indian nationals killed in the West Asia conflict, a police encounter, a family shooting.

Reviews are data. Analyst corrections are stored next to — never over — the model's prediction; "not defence news" verdicts are labelled negatives for a learned relevance filter.

Code: `ingestion/`, `live/`, `llm/explain.py`, `storage/`, `api/live.py`, `dashboard/`.

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5 · Limitations

1. **The labels are the ceiling.** 74.3% narrative agreement; the model is a coin-flip where the annotators disagree. Every score here is bounded by that.
2. **Rare classes are barely measured.** Investigation and Other have 18 and 26 examples in total and 2 each in the held-out set; their held-out F1 of 0.000 rests on four articles.
3. **Severity mixes two meanings.** "Significant topic" and "incident now" share one label, which is why live alerts need source rules.
4. **Headline-only classification is weak and over-confident** (best 0.235 macro-F1). Live, Google decoding stayed rate-limited after the endpoint testing, so in the first hour **64% of classified articles had no text**. The long-run decode rate is unknown.
5. **Propaganda is not assessed.** 9 positives after re-annotation; one reviewer.
6. **The relevance filter is a word list.** Its 99.2% recall is measured on a keyword-collected corpus (an upper bound); by eye about a quarter of what it admits is out of scope.
7. **Fragile and grey dependencies.** The Google News decode uses an undocumented interface; collection scrapes publisher pages. Fine for a personal research tool, not for a public service.
8. **Multilingual capability is untested.** 32 non-English articles; mmBERT's multilinguality is inherited from pre-training, not demonstrated here.
9. **No live ground truth.** The live system's quality is judged by eye and by review clicks; drift over weeks is unmeasured.
10. **LLM summaries are not verified** — only their quotes are. The dashboard labels them as LLM output.

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6 · What would move the numbers

change	expected effect	cost
Adjudicate the ~26% of articles the annotators disagreed on	raises the ceiling every method is under	~280 articles of careful review
Split severity into "incident now" and "strategic significance"	alerts without source rules	relabel 1,064 articles (or the High subset)
Learn the relevance filter from dashboard "not defence news" verdicts	fewer junk articles, measurable precision	a few hundred review clicks — collected as a side effect
Reliable Google News text (publisher search by headline, or a licensed news API)	moves most articles from headline-only to bodied	engineering; possibly a paid API
Propaganda training set	a real propaganda classifier	~600 more random articles reviewed (~8–10 h)
More rare-class examples from the review queue	Investigation / Other become measurable	ongoing review

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7 · Reproduce

```
python -m venv .venv ; .\.venv\Scripts\Activate.ps1
pip install torch --index-url https://download.pytorch.org/whl/cu128
pip install -e ".[dev,serve,ingest,llm]"
pytest                                        # 326 tests

python scripts/01_check_env.py                # GPU, models, Ollama
python scripts/02_build_folds.py --mode both  # dedupe, hold-out, folds   → docs/DATA_REPORT.md
python scripts/03_baselines.py                # §3.1
python scripts/05_train_cv.py --tag m6-e6     # §3.2 (configs/mmbert_small.yaml)
python scripts/07_final_eval.py               # §3.3 — runs once; reprints thereafter
python scripts/07b_holdout_ci.py              # bootstrap intervals
python scripts/07c_headline_routing.py        # §3.5
python scripts/09_inference_check.py          # §3.7
python scripts/10_serve.py --start            # §4: API + collection
cd dashboard ; npm install ; npm run dev      # dashboard → http://localhost:5173
python scripts/12_build_report.py             # this report's figures and PDF
```

The corpus is not in the repository. Hardware: RTX 3050 Laptop 6 GB, Ryzen 5 7235HS, 11.7 GB RAM, Windows 11.

Appendix · decision index

D1 hold out `test.csv`	D2 train on bodied articles	D3 propaganda stays a task	D4 keywords as model input
D5 group-aware folds	D6 five narrative classes	D7 local GPU training	D8 schedule
D9 frozen embeddings + SDPA	D10 early-stopping monitor	D11 the three M5 defects	D12 keyword fusion kept
D13 fixed schedule	D14 512 tokens	D15 final model	D16 held-out result
D17 headline-only routing	D18 propaganda protocol	D19 LLM screener dropped	D20 propaganda review
D21 serving	D22 ingestion + relevance	D23 live pipeline + alert policy	D24 dashboard API
D25 dashboard			

All in DECISIONS.md, each with the evidence that decided it.